

```
[110]: !mamba install pandas matplotlib numpy seaborn
```

```
mambajs 0.21.1
```

```
Specs: xeus-python, numpy, matplotlib, pillow, ipywidgets>=8.1.6, ipyleaflet, scipy, pandas, seaborn
Channels: emscripten-forge-4x, conda-forge
```

```
Solving environment...
Solving took 0.8326999999992549 seconds
All requested packages already installed.
```

```
[111]: from pathlib import Path
import os
```

```
NOTEBOOK_DIR = Path.cwd()
PROJECT_ROOT = NOTEBOOK_DIR.parent
```

```
raw_data_path = PROJECT_ROOT / "data" / "raw"
processed_data_path = PROJECT_ROOT / "data" / "processed"
fig_path = PROJECT_ROOT / "reports" / "figures"
```

```
print("Current directory:", NOTEBOOK_DIR)
print("Project root:", PROJECT_ROOT)
print("Raw data exists:", raw_data_path.exists())
print("Processed data exists:", processed_data_path.exists())
print("Figures folder exists:", fig_path.exists())
```

```
Current directory: /drive/notebooks/cind-820-learning-analytics/notebooks
Project root: /drive/notebooks/cind-820-learning-analytics
Raw data exists: True
Processed data exists: True
Figures folder exists: True
```

```
[112]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[113]: import os
```

```
print("Current working directory:")
print(os.getcwd())
```

```
Current working directory:
/drive/notebooks/cind-820-learning-analytics/notebooks
```

```
[114]: print("\nContents of current folder:")
print(os.listdir("."))
```

```
print("\nContents of parent folder:")
print(os.listdir(".."))
```

```
Contents of current folder:
['oulad_eda.ipynb', 'oulad_modelling.ipynb']
```

```
Contents of parent folder:
['README.md', 'data', 'eda', 'notebooks', 'reports', 'requirements.txt', 'scripts']
```

```
[115]: student_info = pd.read_csv(raw_data_path / "studentInfo.csv")
student_assessment = pd.read_csv(raw_data_path / "studentAssessment.csv")
assessments = pd.read_csv(raw_data_path / "assessments.csv")
vle = pd.read_csv(raw_data_path / "vle.csv")
courses = pd.read_csv(raw_data_path / "courses.csv")
student_registration = pd.read_csv(raw_data_path / "studentRegistration.csv")
```

```
[116]: datasets = {
    "student_info": student_info,
    "student_assessment": student_assessment,
    "assessments": assessments,
    "vle": vle,
    "courses": courses,
    "student_registration": student_registration
}

for name, df in datasets.items():
    print(f"\n{name}")
    print("shape:", df.shape)
    print(df.head(2))
```

```
student_info
shape: (32593, 12)
   code_module code_presentation id_student gender region \
0      AAA      2013J      11391      M      East Anglian Region
1      AAA      2013J      28400      F      Scotland

   highest_education imd_band age_band num_of_prev_attempts studied_credits \
0      HE Qualification 90-100% 55<= 0 240
1      HE Qualification 20-30% 35-55 0 60

   disability final_result
0      N      Pass
1      N      Pass

student_assessment
shape: (173912, 5)
   id_assessment id_student date_submitted is_banked score
0      1752      11391      18      0      78.0
1      1752      28400      22      0      70.0

assessments
shape: (206, 6)
   code_module code_presentation id_assessment assessment_type date weight
0      AAA      2013J      1752      TMA      19.0      10.0
1      AAA      2013J      1752      TMA      54.0      20.0
```

```

vle
shape: (6364, 6)
   id_site code_module code_presentation activity_type week_from week_to
0   546943         AAA         2013J         resource         NaN         NaN
1   546712         AAA         2013J         oucontent         NaN         NaN

courses
shape: (22, 3)
   code_module code_presentation module_presentation_length
0         AAA         2013J                268
1         AAA         2014J                269

student_registration
shape: (32593, 5)
   code_module code_presentation id_student date_registration \
0         AAA         2013J         11391         -159.0
1         AAA         2013J         28400         -53.0

   date_unregistration
0                 NaN
1                 NaN

```

```

[117]: assessment_merged = student_assessment.merge(
        assessments,
        on="id_assessment",
        how="left"
    )

    assessment_features = assessment_merged.groupby(
        ["id_student", "code_module", "code_presentation"],
        as_index=False
    ).agg(
        mean_score=("score", "mean"),
        total_assessments=("id_assessment", "count"),
        mean_weight=("weight", "mean")
    )

    assessment_features.head()

```

```

[112]:
   id_student code_module code_presentation mean_score total_assessments mean_weight
0         6516         AAA         2014J         61.800000             5         20.000000
1         8462         DDD         2013J         87.666667             3         13.333333
2         8462         DDD         2014J         86.500000             4         12.500000
3        11391         AAA         2013J         82.000000             5         20.000000
4        23629         BBB         2013B         82.500000             4          6.250000

```

```

[118]: df = student_info.merge(
        assessment_features,
        on=["id_student", "code_module", "code_presentation"],
        how="left"
    )

    print(df.shape)
    df.head()

```

```

(32593, 15)
[113]:
   code_module code_presentation id_student gender region highest_education imd_band age_band num_of_prev_attempts studied_credits disability
0         AAA         2013J         11391      M  East Anglian Region  HE Qualification  90-100%  55<=             0             240          N
1         AAA         2013J         28400      F  Scotland  HE Qualification  20-30%  35-55             0             60          N
2         AAA         2013J         30268      F  North Western Region  A Level or Equivalent  30-40%  35-55             0             60          Y
3         AAA         2013J         31604      F  South East Region  A Level or Equivalent  50-60%  35-55             0             60          N
4         AAA         2013J         32885      F  West Midlands Region  Lower Than A Level  50-60%  0-35             0             60          N

```

```

[119]: print(df.columns.tolist())
        print(df["final_result"].value_counts(dropna=False))
        print(df[["mean_score", "total_assessments", "mean_weight"]].isna().sum())

['code_module', 'code_presentation', 'id_student', 'gender', 'region', 'highest_education', 'imd_band', 'age_band', 'num_of_prev_attempts',
'studied_credits', 'disability', 'final_result', 'mean_score', 'total_assessments', 'mean_weight']
final_result
Pass          12361
Withdrawn     10156
Fail          7052
Distinction   3024
Name: count, dtype: int64
mean_score    6773
total_assessments 6750
mean_weight    6750
dtype: int64

```

```

[120]: df.to_csv(processed_data_path / "oulad_student_level.csv", index=False)

```

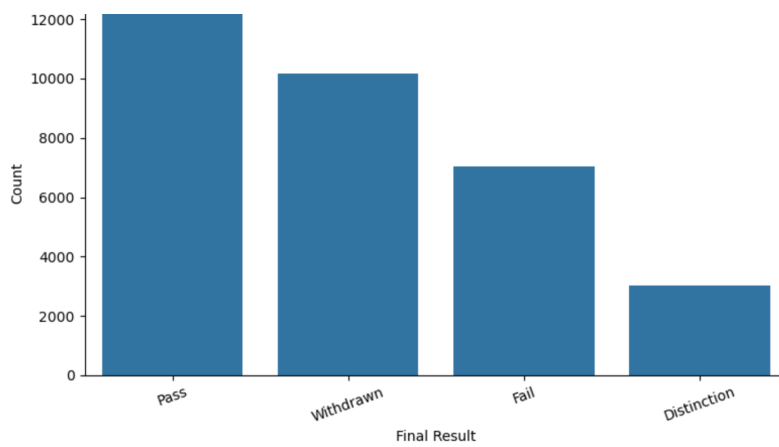
```

[121]: import matplotlib.pyplot as plt
        import seaborn as sns

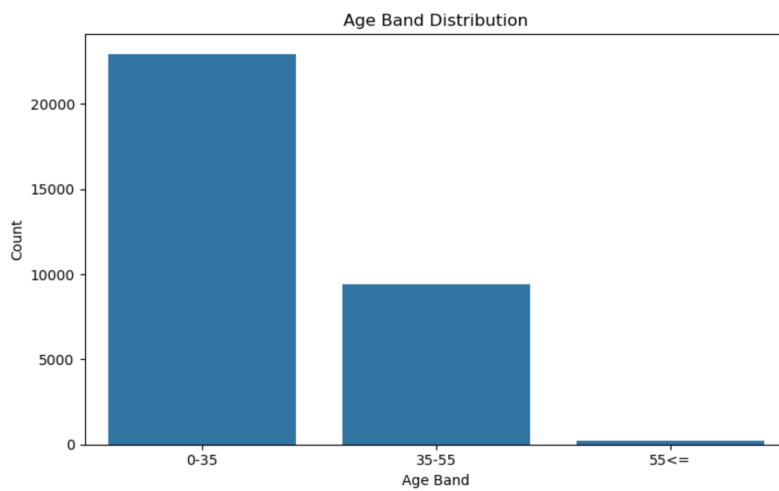
        plt.figure(figsize=(8,5))
        sns.countplot(data=df, x="final_result", order=df["final_result"].value_counts().index)
        plt.title("Final Result Distribution")
        plt.xlabel("Final Result")
        plt.ylabel("Count")
        plt.xticks(rotation=20)
        plt.tight_layout()
        plt.savefig(fig_path / "correlation_heatmap.png")
        plt.show()

```

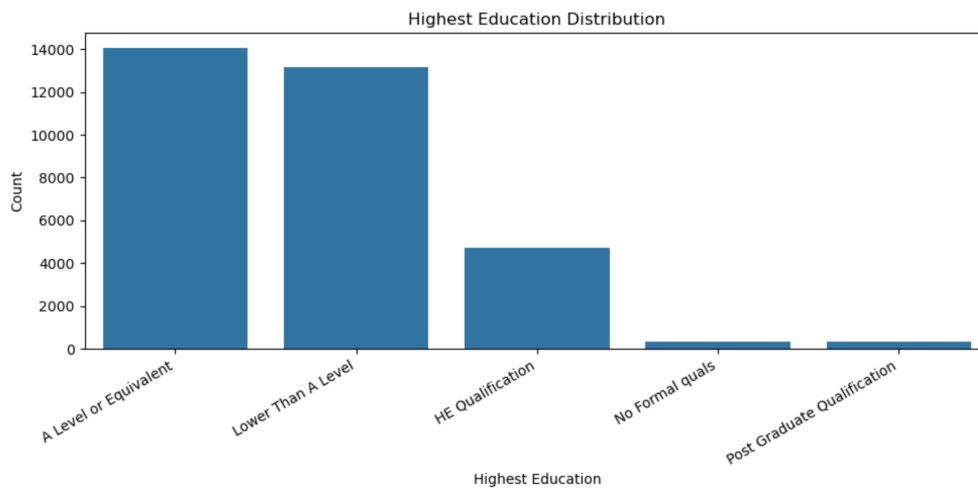
Final Result Distribution



```
[122]: plt.figure(figsize=(8,5))
sns.countplot(data=df, x="age_band", order=df["age_band"].value_counts().index)
plt.title("Age Band Distribution")
plt.xlabel("Age Band")
plt.ylabel("Count")
plt.tight_layout()
plt.savefig(fig_path / "age_band_distribution.png")
plt.show()
```

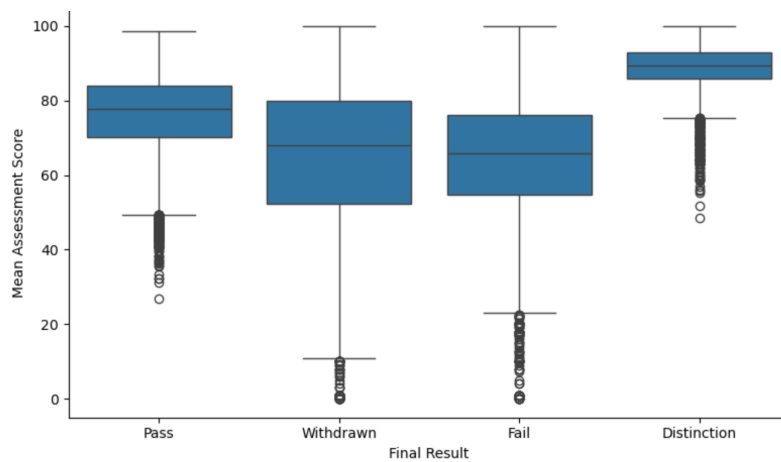


```
[123]: plt.figure(figsize=(10,5))
sns.countplot(data=df, x="highest_education", order=df["highest_education"].value_counts().index)
plt.title("Highest Education Distribution")
plt.xlabel("Highest Education")
plt.ylabel("Count")
plt.xticks(rotation=30, ha="right")
plt.tight_layout()
plt.savefig(fig_path / "highest_education_distribution.png")
plt.show()
```



```
[124]: plt.figure(figsize=(8,5))
sns.boxplot(data=df, x="final_result", y="mean_score")
plt.title("Mean Assessment Score by Final Result")
plt.xlabel("Final Result")
plt.ylabel("Mean Assessment Score")
plt.tight_layout()
plt.savefig(fig_path / "mean_score_by_final_result.png")
plt.show()
```

Mean Assessment Score by Final Result



```
[125]: numeric_cols = [
        "num_of_prev_attempts",
        "studied_credits",
        "mean_score",
        "total_assessments",
        "mean_weight"
    ]

    corr = df[numeric_cols].corr()
    corr.to_csv(processed_data_path / "correlation_matrix.csv")
    corr

    plt.figure(figsize=(8,6))
    sns.heatmap(corr, annot=True, cmap="Blues", fmt=".2f")
    plt.title("Correlation Heatmap of Numeric Features")
    plt.tight_layout()
    plt.savefig(fig_path / "correlation_heatmap.png")
    plt.show()
```



```
[126]: print("Shape:", df.shape)
        print("\nColumns:")
        print(df.columns.tolist())

        print("\nData types:")
        print(df.dtypes)

        print("\nPreview:")
        display(df.head())

        Shape: (32593, 15)

        Columns:
        ['code_module', 'code_presentation', 'id_student', 'gender', 'region', 'highest_education', 'imd_band', 'age_band', 'num_of_prev_attempts',
        'studied_credits', 'disability', 'final_result', 'mean_score', 'total_assessments', 'mean_weight']

        Data types:
        code_module          str
        code_presentation    str
        id_student           int64
        gender               str
        region               str
        highest_education    str
        imd_band             str
        age_band             str
        num_of_prev_attempts int64
        studied_credits      int64
        disability           str
        final_result         str
        mean_score           float64
        total_assessments    float64
        mean_weight          float64
```

dtype: object

Preview:

	code_module	code_presentation	id_student	gender	region	highest_education	imd_band	age_band	num_of_prev_attempts	studied_credits	disability
0	AAA	2013J	11391	M	East Anglian Region	HE Qualification	90-100%	55<=	0	240	N
1	AAA	2013J	28400	F	Scotland	HE Qualification	20-30%	35-55	0	60	N
2	AAA	2013J	30268	F	North Western Region	A Level or Equivalent	30-40%	35-55	0	60	Y
3	AAA	2013J	31604	F	South East Region	A Level or Equivalent	50-60%	35-55	0	60	N
4	AAA	2013J	32885	F	West Midlands Region	Lower Than A Level	50-60%	0-35	0	60	N

```
[127]: missing_summary = pd.DataFrame({
    "missing_count": df.isna().sum(),
    "missing_percent": (df.isna().sum() / len(df) * 100).round(2)
}).sort_values("missing_percent", ascending=False)

missing_summary
```

```
[122]:
```

	missing_count	missing_percent
mean_score	6773	20.78
total_assessments	6750	20.71
mean_weight	6750	20.71
imd_band	1111	3.41
code_module	0	0.00
code_presentation	0	0.00
id_student	0	0.00
gender	0	0.00
region	0	0.00
highest_education	0	0.00
age_band	0	0.00
num_of_prev_attempts	0	0.00
studied_credits	0	0.00
disability	0	0.00
final_result	0	0.00

```
[128]: missing_summary.to_csv(processed_data_path / "missing_summary.csv")
```

```
[129]: numeric_summary = df.describe(include=[np.number]).T
numeric_summary
```

```
[124]:
```

	count	mean	std	min	25%	50%	75%	max
id_student	32593.0	706687.669131	549167.313855	3733.0	508573.000000	590310.0	644453.000000	2716795.0
num_of_prev_attempts	32593.0	0.163225	0.479758	0.0	0.000000	0.0	0.000000	6.0
studied_credits	32593.0	79.758691	41.071900	30.0	60.000000	60.0	120.000000	655.0
mean_score	25820.0	72.768254	16.374997	0.0	64.666667	76.0	84.545455	100.0
total_assessments	25843.0	6.729559	3.771381	1.0	4.000000	7.0	10.000000	14.0
mean_weight	25843.0	13.350761	8.823415	0.0	8.333333	9.9	20.000000	100.0

```
[130]: numeric_summary.to_csv(processed_data_path / "numeric_summary.csv")
```

```
[131]: categorical_vars = ["final_result", "age_band", "highest_education", "gender", "region"]
```

```
for col in categorical_vars:
    print(f"\n--- {col} ---")
    print(df[col].value_counts(dropna=False))
```

```
--- final_result ---
final_result
Pass      12361
Withdrawn 10156
Fail       7052
Distinction 3024
Name: count, dtype: int64
```

```
--- age_band ---
age_band
0-35      22944
35-55     9433
55<=       216
Name: count, dtype: int64
```

```
--- highest_education ---
highest_education
A Level or Equivalent      14045
Lower Than A Level        13158
HE Qualification           4730
No Formal quals            347
Post Graduate Qualification 313
Name: count, dtype: int64
```

```
--- gender ---
gender
M      17875
F      14718
Name: count, dtype: int64
```

```
--- region ---
region
Scotland      3446
```

```

East Anglian Region    3340
London Region          3216
South Region           3092
North Western Region   2906
West Midlands Region   2582
South West Region       2436
East Midlands Region    2365
South East Region       2111
Wales                  2086
Yorkshire Region        2006
North Region            1823
Ireland                 1184
Name: count, dtype: int64

```

```

[132]: for col in categorical_vars:
        freq = df[col].value_counts(dropna=False).reset_index()
        freq.columns = [col, "count"]
        freq.to_csv(processed_data_path / f"{col}_frequency.csv", index=False)

```

```

[133]: dup_count = df.duplicated(subset=["id_student", "code_module", "code_presentation"]).sum()
        print("Duplicate student-module-presentation rows:", dup_count)

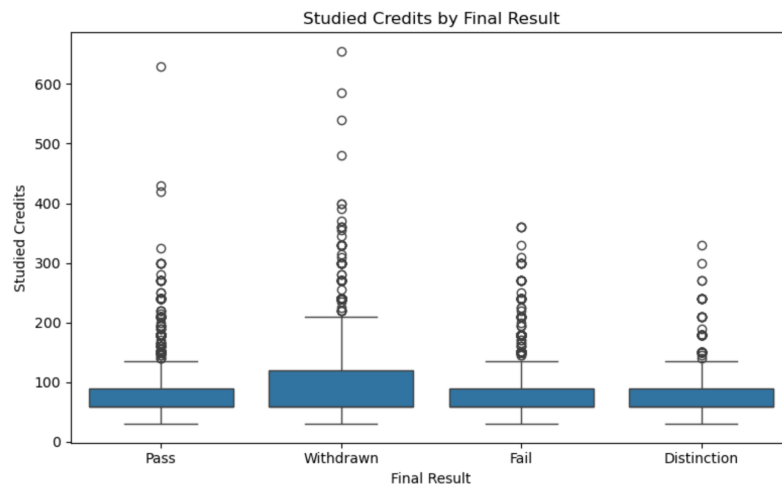
Duplicate student-module-presentation rows: 0

```

```

[134]: plt.figure(figsize=(8,5))
        sns.boxplot(data=df, x="final_result", y="studied_credits")
        plt.title("Studied Credits by Final Result")
        plt.xlabel("Final Result")
        plt.ylabel("Studied Credits")
        plt.tight_layout()
        plt.savefig(fig_path / "studied_credits_by_final_result.png", dpi=300, bbox_inches="tight")
        plt.show()

```



```

[135]: df.to_csv(processed_data_path / "oulad_student_level.csv", index=False)

```

```

[ ]:

```